

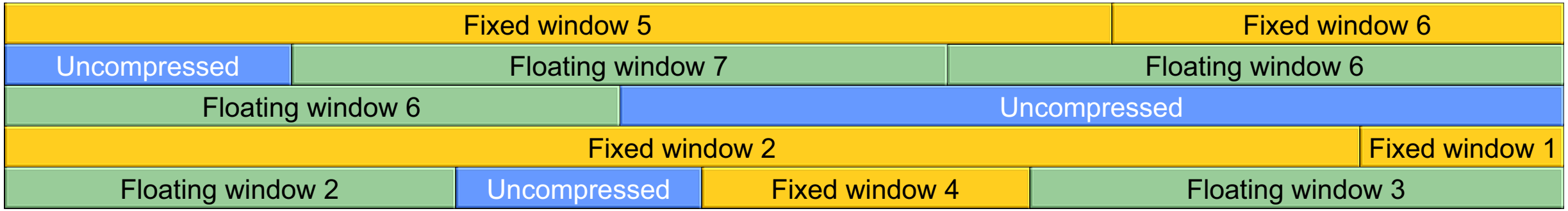
Dmitry Muravyev

Delta encoding

Sprint 2: Building a chain of blocks

Chain of blocks

- The main and very difficult task is to build as efficient a sequence of blocks as possible.

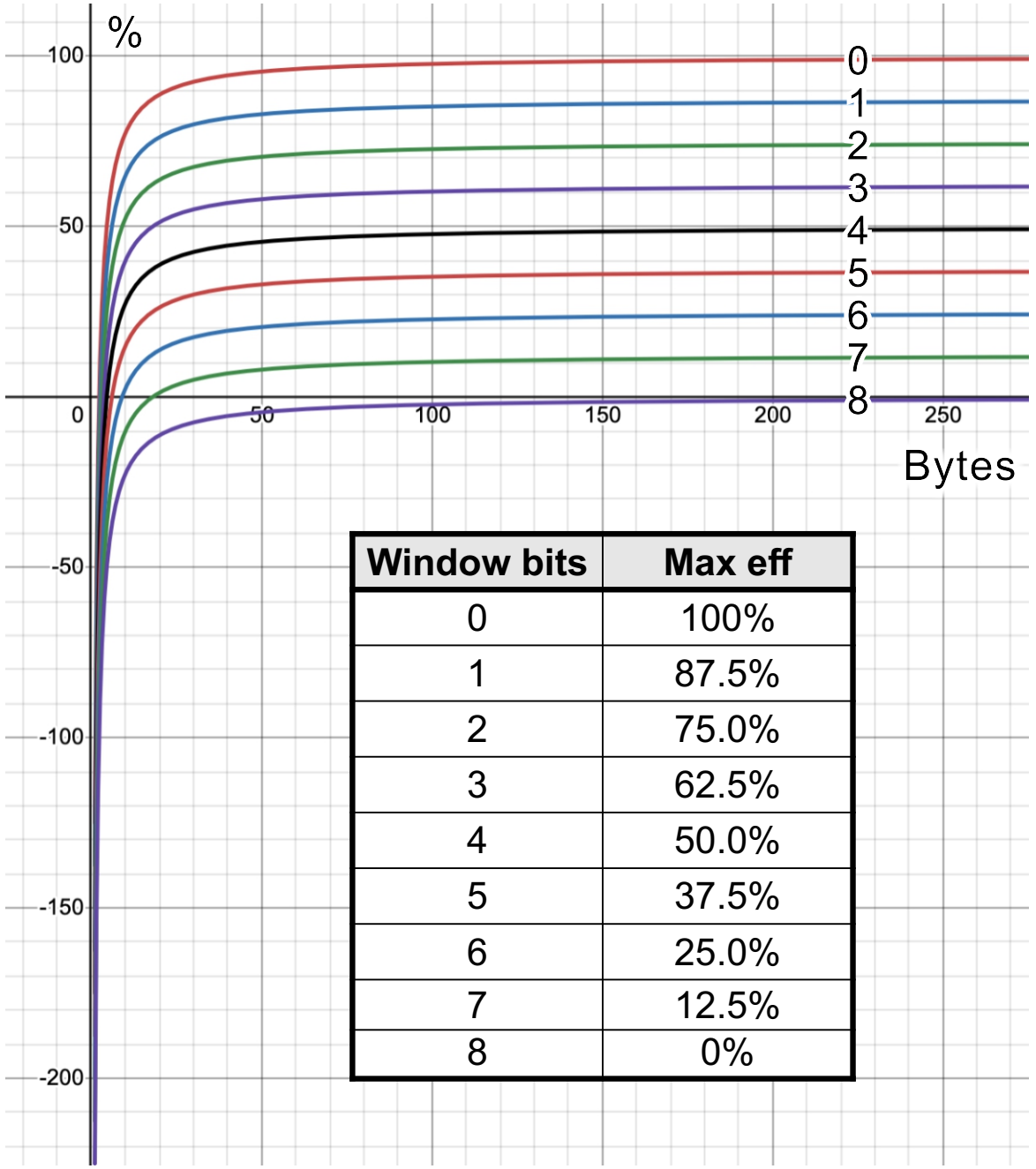


Efficiency

$E_{block} = (1 - B / R_{block}) \times 100\%,$
$E_{overall} = (1 - C / R) \times 100\%,$

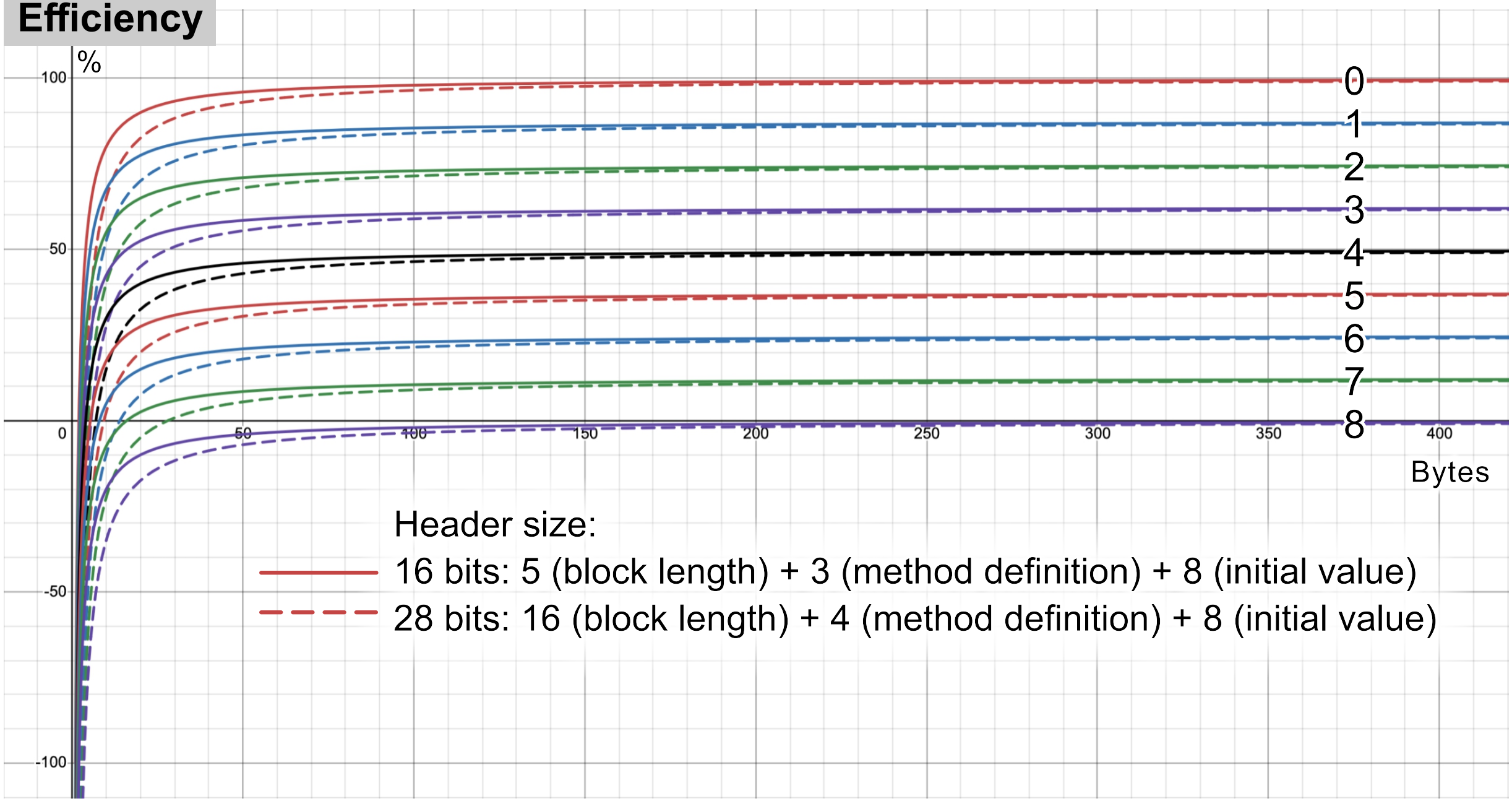
- B — Block size, B = Header size + Method (m) × Block frames,
- R_{block} — Raw (uncompressed) data size per block,
- R — Overall raw data size,
- C — Compressed data size (sum of all block sizes).

* All sizes are in bits.

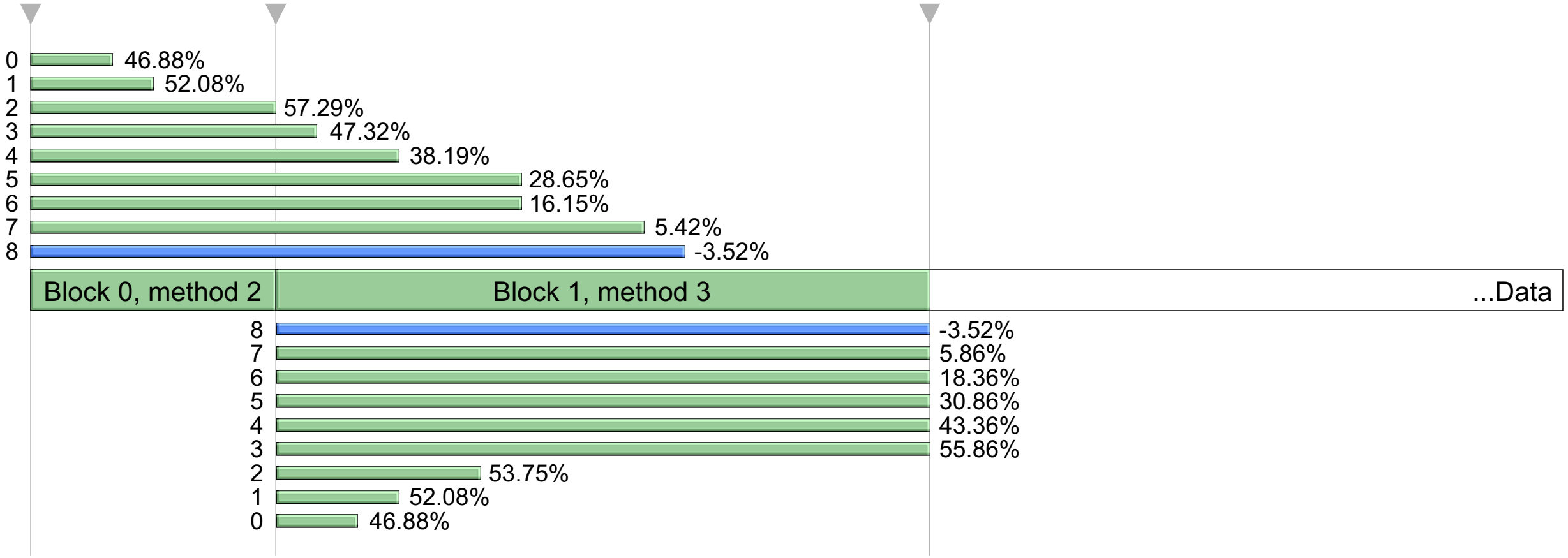


Window bits	Max eff
0	100%
1	87.5%
2	75.0%
3	62.5%
4	50.0%
5	37.5%
6	25.0%
7	12.5%
8	0%

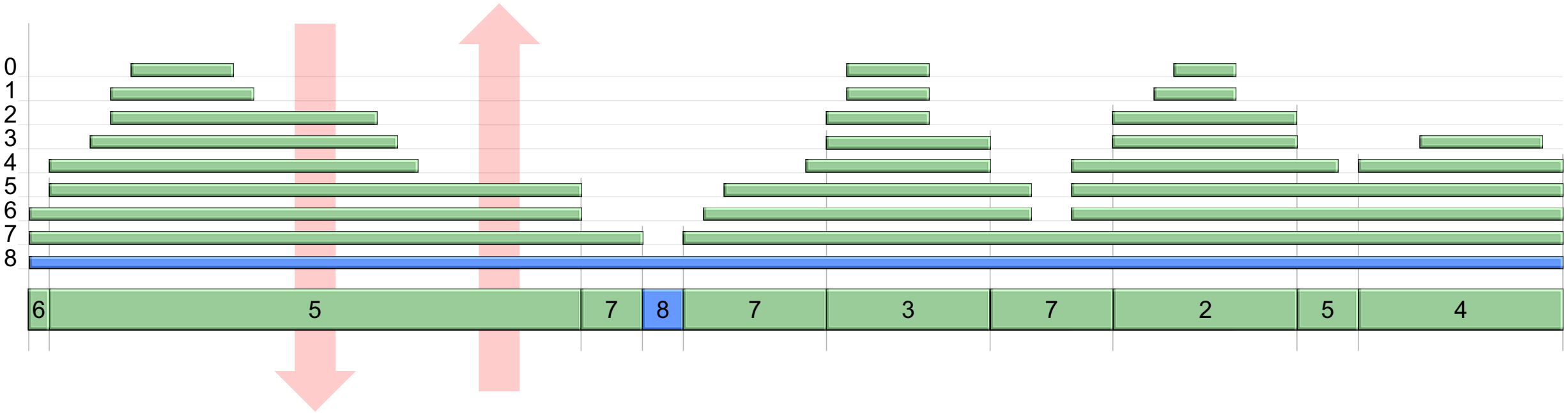
Efficiency


%
100
50
0
-50
-100
50
100
150
200
250
300
350
400
Bytes
0
1
2
3
4
5
6
7
8
Header size:
16 bits: 5 (block length) + 3 (method definition) + 8 (initial value)
28 bits: 16 (block length) + 4 (method definition) + 8 (initial value)

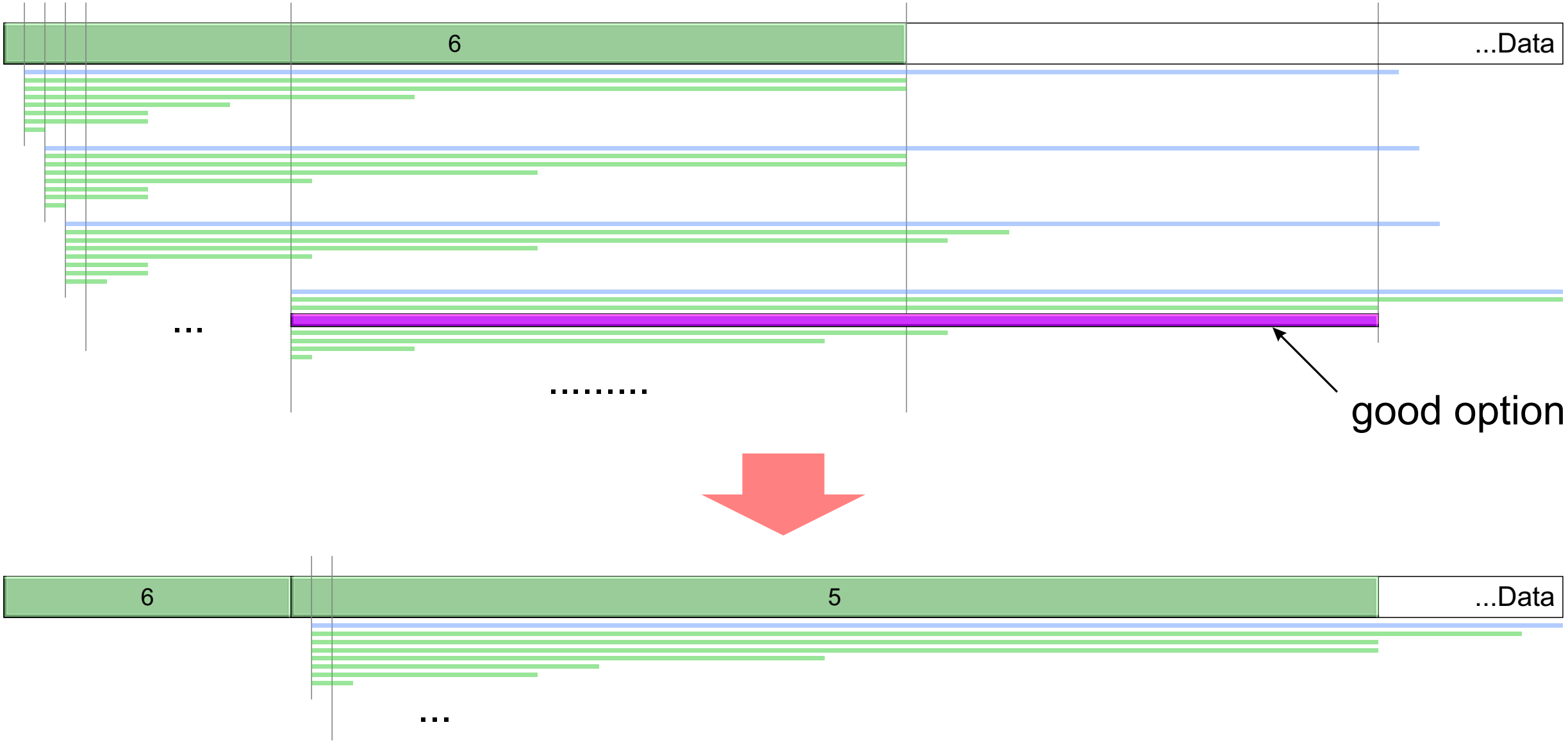
Sequential block search



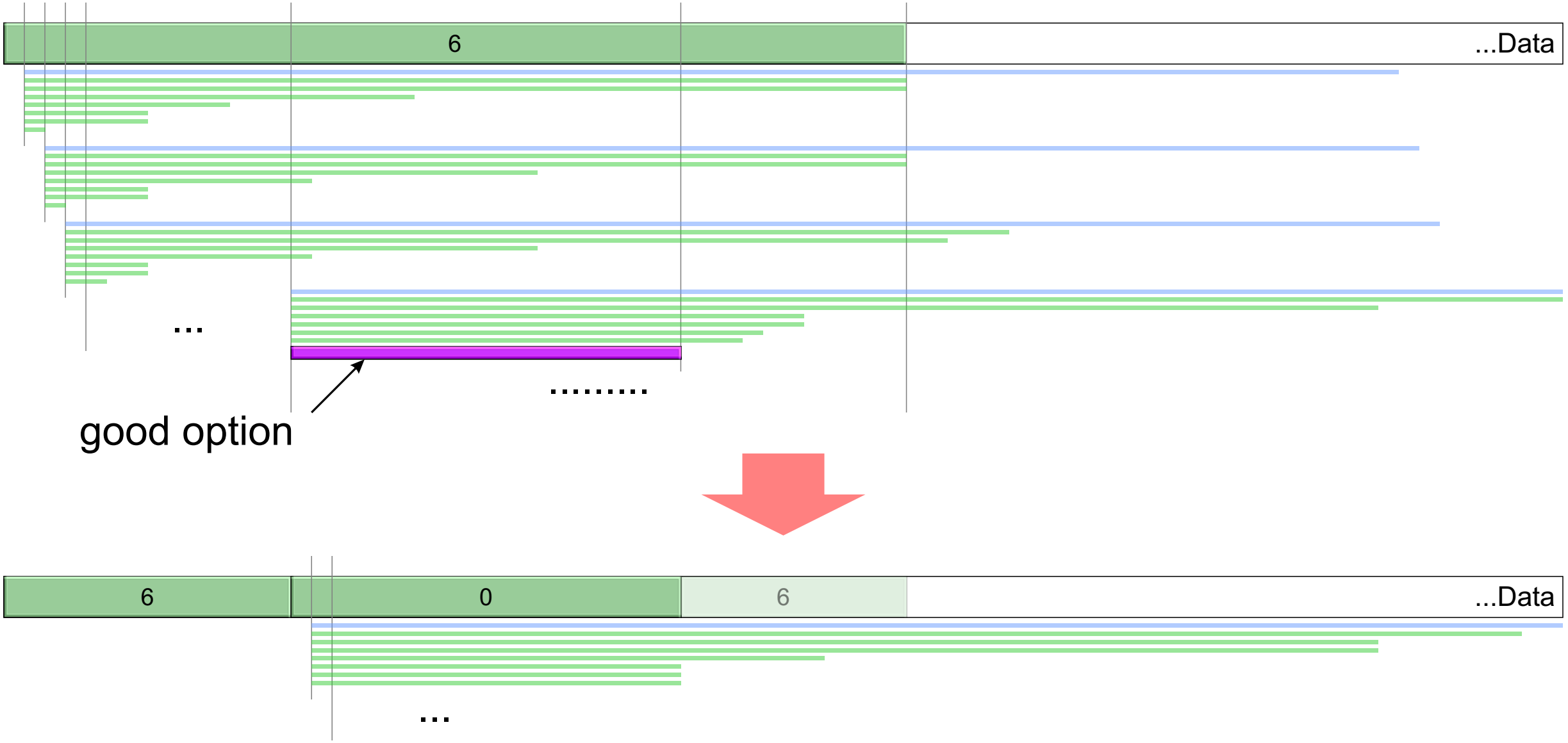
Method map (concept)



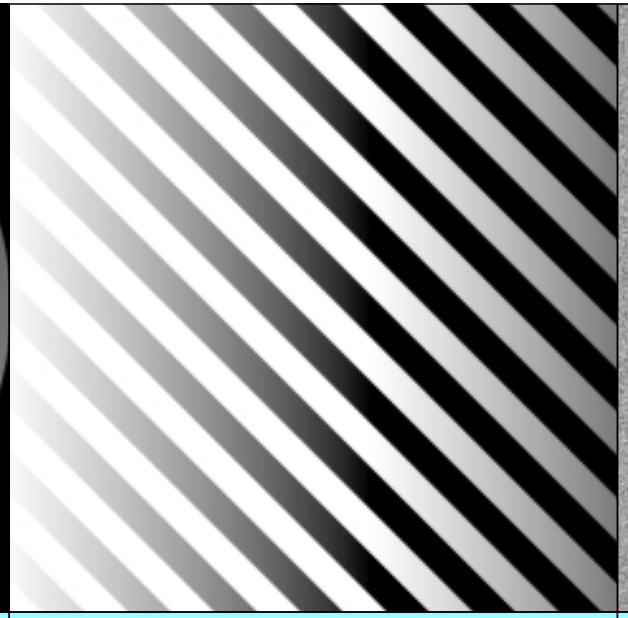
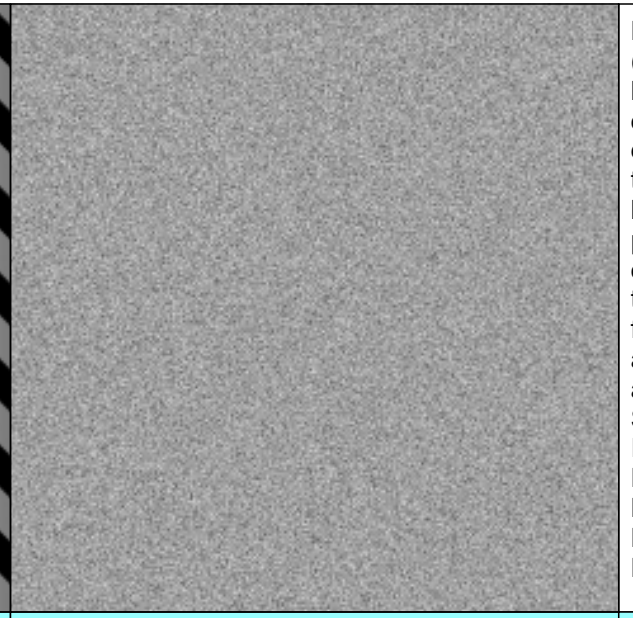
Frame-by-frame efficiency analysis



Frame-by-frame efficiency analysis



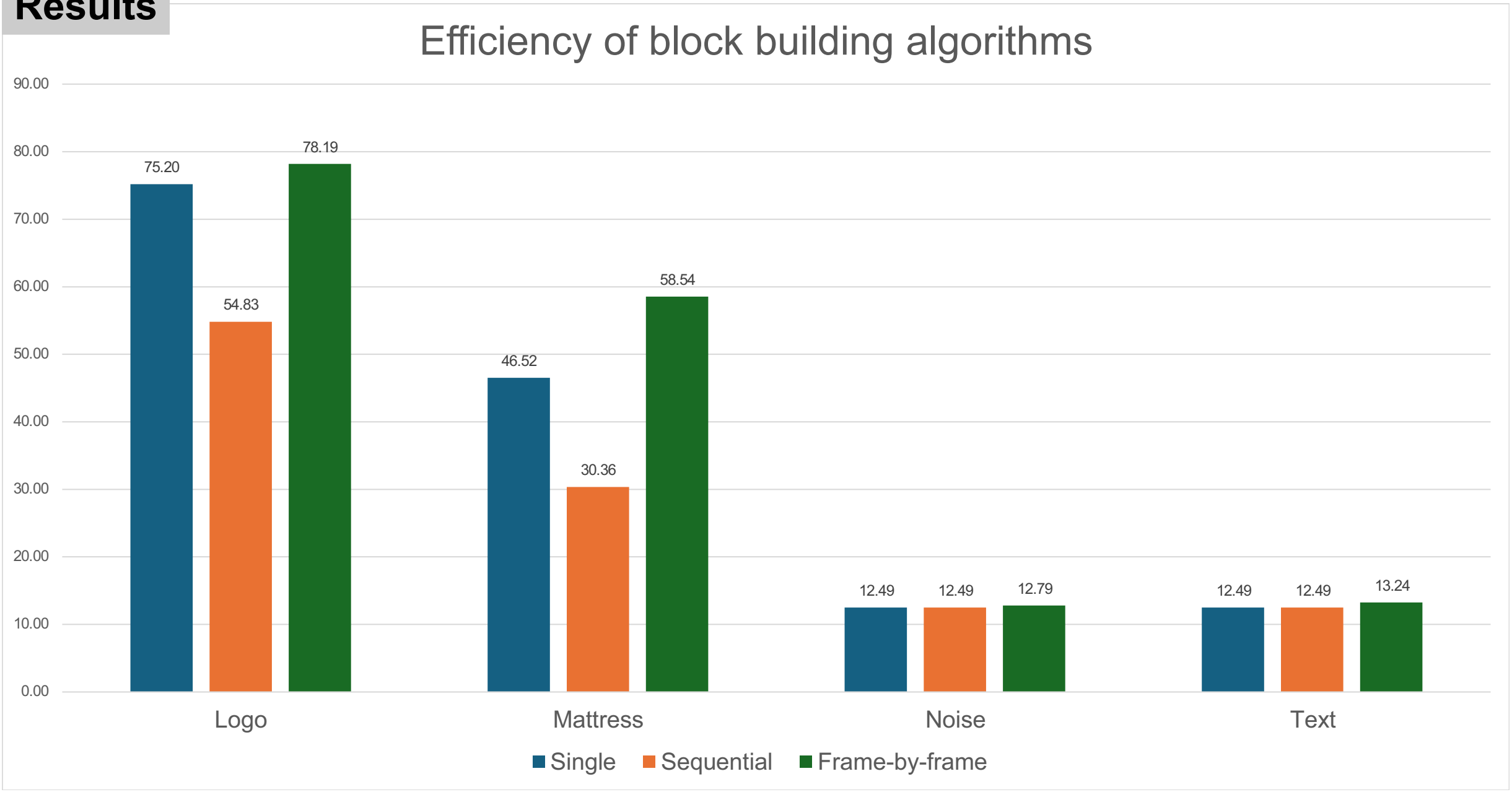
Results

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6 bits, 75.20%	4 bits, 46.52%	16 bits, 12.49%	15 bits, 12.49%
3 bits, 54.83%	3 bits, 30.36%	16 bits, 12.49%	15 bits, 12.49%
7 bits, 78.19%	4 bits, 58.54%	10 bits, 12.79%	15 bits, 13.24%

Block size field (bits) / The algorithm for building blocks:

Single method
Sequential search
Frame-by-frame efficiency analysis

Results


Efficiency of block building algorithms
90.00
80.00
70.00
60.00
50.00
40.00
30.00
20.00
10.00
0.00
75.20
54.83
78.19
46.52
30.36
58.54
12.49
12.49
12.79
12.49
12.49
13.24
Logo
Mattress
Noise
Text
Single
Sequential
Frame-by-frame

Next sprint

- Channels: RGB, ARGB, L/R.
- Splitting images into squares.
- Header optimization.

